

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Data Communication (New)

Semester: Spring

Year : 2025
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What is data communication system? Explain the block diagram of data communication. 7
b) Define system. Explain basic system properties with its application and example. 8
2. a) Explain different types of elementary signals with its waveform and mathematical representation. 8
b) Explain the operational principles of the ALOHA protocol in data communication. Compare it with CSMA/CD in terms of collision handling. 7
3. a) Define UTP & STP. Explain the working principle of optical fiber. Discuss the advantages and disadvantages of fiber optics. 8
b) Explain various transmission impairments of a communication system. 7
4. a) How collision can be detected and corrected in data transmission? Explain brief the flow chart about CSMA/CD. 8

OR

A bit stream 11010101 is transmitted using standard CRC method. The generator polynomial is x^3+1 . Show the actual bit transmitted. Suppose, the third bit from the left is inverted during transmission. Show that this error is detected at the receiver's end.

- b) What is ARQ? How does stop and wait ARQ and sliding window ARQ deals with error. 7

OR

Describe flow control and error control. How does Go-back -N ARQ work?

5. a) What do you understand by PBX? Compare among circuit switching, message switching and packet switching. 8

- b) Explain FDM. State sampling theorem and why we need modulation in communication system? 7
- 6. a) Define line coding. Explain the block diagram of PCM system. 7
- b) Encode digital data 11010101011101 using 8
 - i) Unipolar RZ ii) Bipolar NRZ iii) AMI iv) Manchester
- 7. Write short notes on: (**Any two**) 2×5
 - a) Shannon Capacity Theorem
 - b) Lossy compression
 - c) Point-to-Point protocol

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1. a) Explain the five essential components of a data communication system. How does each component contribute to successful communication? Explain with suitable diagram. 7
- b) What is serial and parallel data transmission? Differentiate between synchronous and asynchronous data transmission. 8
2. a) Check whether the given system is linear, time invariant and causal or not. 7
 - i. $y(t) = \log x(t)$
 - ii. $y(t) = t^2 x(t)$
- b) Differentiate between energy and power signal. Calculate whether ramp signal is power signal or energy. 8
3. a) Why Frame Relay is faster than X.25? Explain TCP/IP layers and compare it with OSI reference model. 8
- b) What is the purpose of cladding in an optical fibre? Explain the modes of propagation used in the optical fibre. 7
4. a) Define transmission impairments. And explain the causes of transmission impairments in details. 7
- b) If the 7 bit Hamming codeword received by a receiver is 1011011. Assuming the even parity, state whether the received codeword is correct or wrong. If wrong, decode the correct data word and locate the bit in error. 8

OR

Compress “Engineering” using Huffman codes and find its efficiency and redundancy.

5. a) What is ARQ? What are the two types of sliding window ARQ error control? Explain. 8

- b) What do you mean by multiplexing? How is TDM different from FDM? 7
6. a) Why do we need modulation need in data communication? Explain. 7

OR

Calculate the rate of a 500 baud 8-QAM signal with a constellation diagram.

- b) Represent the given bit stream 10011100001101 using 8
- i. Polar NRZ
 - ii. Differential Manchester
 - iii. HDB3
7. Write short notes on: (**Any two**) 2×5
- a) Standards organizations
 - b) Noise and its types
 - c) HDLC protocol vs Point-to-Point protocol

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1. a) What is the role of standard organizations in data communication? 8
Differentiate between analog and digital data communication system.
- b) Explain Shannon's theorem for a noisy channel. Find the maximum 7
bit rate for a channel having bandwidth 3100 Hz and SNR ratio of 20 dB.
2. a) Define continuous and discrete time signals. Illustrate graphically 7
unit step signal, ramp signal, impulse signal, sinusoidal signal and signum signal with their mathematical relation.
- b) Define LTI system. Check the linearity and time invariance of the 8
following system.
$$y(n) = nx(n)$$
3. a) Explain layers of the OSI model. 7
- b) Define electromagnetic spectrum for telecommunication. Briefly 8
explain ground wave, sky wave and line of sight (LOS) propagation.
4. a) Briefly explain attenuation, distortion and noise that occurs in data 7
communication. Why are they called transmission impairments?
- b) If the frame is $x^7 + x^5 + 1$ by the generator polynomial $x^3 + 1$, what 8
would be the transmitted frame using CRC?

OR

Encode the given five source symbols (messages) using Huffman codes and find its efficiency

Message	m_1	m_2	m_3	m_4	m_5
Probability	0.4	0.2	0.2	0.1	0.1

5. a) When is the Retransmission necessary? Explain about stop and wait 7
ARQ protocol for the damaged frame.
- b) Briefly explain circuit switching and packet switching. 8

6. a) How does delta modulation overcome the demerits of PCM? Briefly explain the working of delta modulation. 8

OR

Briefly explain the digital modulation techniques with suitable diagram.

- b) What is a modem? Show Manchester encoding pattern for the bit stream 10111001101. 7
7. Write short notes on: (**Any two**) 2×5
- a) Optical fiber
 - b) HDLC protocol
 - c) Run Length Coding

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1. a) Define Simplex, Half Duplex and Full Duplex mode of data transmission. Briefly explain digital communication system with the help of general block diagram. 8
 - b) Differentiate between different types of transmission modes. Briefly Explain RS- 232C. 7
 2. a) Define Signal. Briefly explain periodic and non-periodic signals, energy and power signals, deterministic and random signals, continuous and discrete time signals. 7
 - b) Define unit step signal with necessary mathematical expression and graphical illustration. Check the following system for linearity, causality, static & dynamic and time invariance. 8
- $$y(n) = Ax(n) + B$$
3. a) Compare and contrast between OSI and TCP/IP with necessary figures. 7
 - b) Define guided and unguided transmission media. State the advantages of optical fiber over other guided media. 8
 4. a) Why burst error is difficult to handle than single error? Explain why CRC is better than parity method of error detection. 7

OR

Explain various transmission impairments of a communication system.

- b) Compare and contrast between lossless and lossy compression. 8
5. a) Explain the importance of flow control. Briefly explain sliding window flow control mechanism. 8
- b) Explain the multiplexing technique applied in digital telephony with their significance, application and multiplexing hierarchy. 7
6. a) Is switching and routing same? Differentiate between message switching, circuit switching and packet switching. 8

- b) Differentiate between PSK, ASK and FSK modulation techniques with necessary diagram. 7

OR

Encode the Bit Stream 011010001 using the following scheme.

- i. NRZ – I
- ii. NRZ _ L
- iii. Manchester
- iv. Differential Manchester

7. Write short notes on: (**Any two**) 2×5
- a) Cellular Telephony
 - b) Point to Point protocol
 - c) QPSK